PulseIQ Analytics — Phase 2 Report

Participant: Shreya Raut **Track:** Data Analytics **Phase:** Phase 2

Repository URL: Add your actual GitHub URL before submission **Git tag:** phase-2 (create/push after review)

Task status and evidence

Task	Status	Evidence included
Day 6 — Attendance & leave	Implemented	App page + attendance/leave CSVs
Day 7 — Attrition	Implemented	Actual exits + grouped prediction view
Day 8 — Team productivity	Implemented	Team/sprint selectors + metric rules
Day 9 — Payroll	Partially complete	Cost page + arithmetic checks; official key missing
Day 10 — Budget & KPIs	Implemented	Variance chart/ranking + KPI target traffic lights

Numeric findings from supplied data

• **Attendance:** May 2026 was lowest at **92.64%** (7,551 / 8,151 scheduled days); January 2026 was 93.11% and June 2026 was 93.16%.

• **Department attendance:** Finance & Accounts was **90.60%** (5,505 / 6,076); Data & AI was 94.84% (17,732 / 18,697). Compare working-day mix and context before drawing conclusions.

• **Approved leave:** Earned/Privilege Leave 2,336 days; Casual Leave 1,995 days; Sick Leave 1,769 days.

• **Attrition:** October 2025 recorded 10 exits (9 voluntary); March 2025 recorded 9; December 2023 recorded 8.

• **Team delivery:** Support Tooling logged 10,288 story points, 177 blocked tasks and 142 reopened counts. Sales Enablement logged 8,424 story points, 131 blocked tasks and 150 reopened counts.

• **Payroll budget:** Matched employer cost was ■861,997,360 vs ■862,030,000 budget (net –■32,640). Department-level differences remain: Data & AI +■992,768; Product & Design +■763,164; Customer Support +■482,051.

Validation, privacy and limitations

• Automated test suite: **7 passed**. Payroll arithmetic checks passed within ■0.05 tolerance for all 7,344 rows: gross pay composition, net pay less deductions, and employer cost including employer PF/ESI.

• Two employee-month duplicate pairs were detected and retained for review; no rows were silently deleted. The official **unlock_PQ-09** answer key was not included in the supplied ZIP, so official answer-key reconciliation remains pending.

• The preparation environment did not have Streamlit installed. The app source was syntax-checked and data tests passed, but live browser verification and genuine dashboard screenshots still need to be completed after installing requirements and running `streamlit run app.py`.

• Predicted attrition is labelled “Predicted, not certain” and shown at department level only for groups with at least five predictions. Do not use prediction scores as individual employment decisions.

• KPI status rules: attendance and task completion are green when at/above target; attrition and overtime share are green when at/below target. Overtime share is overtime hours divided by scheduled hours (scheduled days × 8), expressed as a percentage.

What I learned

I built views for attendance, approved leave, actual exits, predicted risk groups, team delivery and payroll cost. I used explicit metric definitions and checked payroll arithmetic rather than assuming totals were correct. I learned that aggregate results can hide department-level differences and that predictions must be separated from actual outcomes.

Problems and resolutions

• Missing official payroll validation key: added arithmetic checks and duplicate-row review; did not claim official reconciliation.

• Sensitive attrition risk data: the risk section avoids employee names/IDs and suppresses groups smaller than five.

• Cancelled tasks: excluded from delivery-rate denominators and shown separately; velocity counts story points only for Done tasks.

Before portal submission

Install requirements, run and verify the app, capture genuine screenshots, complete official payroll answer-key reconciliation if the key is provided, add the real GitHub URL, create and push the `phase-2` tag, and upload this single PDF through the programme site.